① Handle Distributed Transactions | Two-Phase Commit (2PC), Three-Phase Commit (3PC), SAGA Pattern

Lecture 57

What is Transaction :→

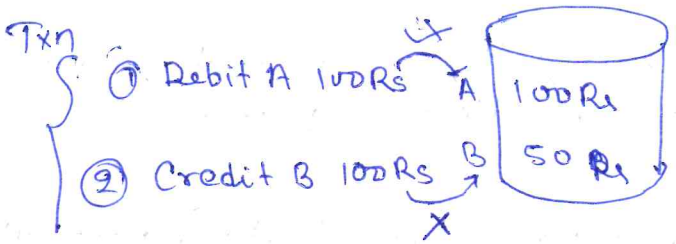
It refers to set of operations which need to be performed or or simply say group of tasks which need to be performed against the DB.

It has 4 properties :-

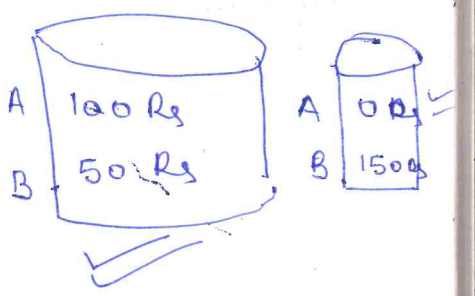
- Atomicity :- All operations in a single transaction should be success or all should fail.
- Consitency :- DB should be consistent state before and after the transaction.
- Isolation :- More than one transaction that is running concurrently, appears to be serialized.
- Durability :- After transaction sucessfully completed even if DB fails, data should not get lost.

Concept&&Coding

Atomicity

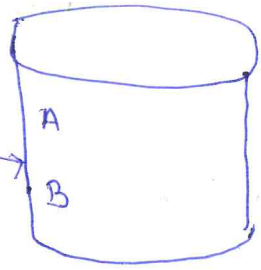


Consitency



Isolation

Txn 1
{
Debit A 100Rs
Credit B 50Rs
}

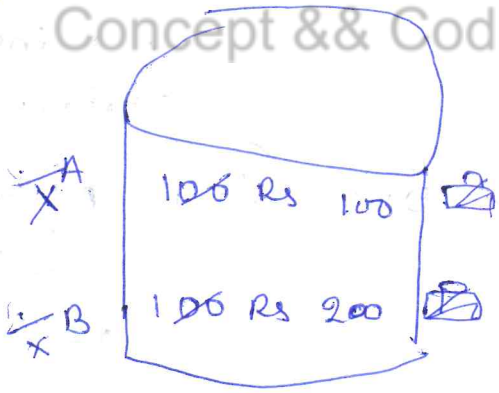


Txn 2
{
Debit B 150Rs
Credit C 150Rs
}

Txn
Begin_Transaction: (T) DB
- Withdrawal Money from 'A': Rs100
- Deposit Money to 'B': Rs 100

DB : T

if all success:
Do Commit
else
Do Rollback
Close_Transaction;



Concept && Coding

How to handle this Distributed System, where operations involves multiple databases?

or

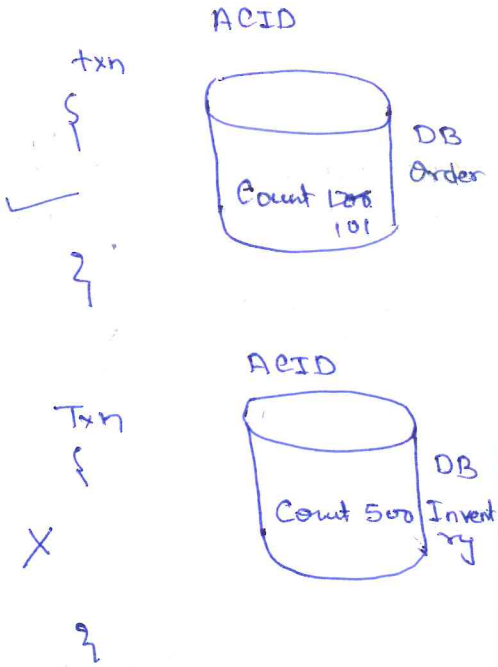
As Transaction is Local to a particular database? How will satisfy the transaction Property in Distributed system.

```
Begin_Transaction
    - update Order DB
    - update Inventory DB

if all sucess:
        Do Commit
else
        Do Rollback

Close_Transaction;
```

Will this work ??



Concept && Coding

How to handle Transaction in Distributed System

① **2-Phase Commit** (very popular) (Synchornious)

② **3-Phase Commit** (No Much used because of Complexity) (Synchornious)

③ **Saga Pattern** (still used very much) (Asynchornious)

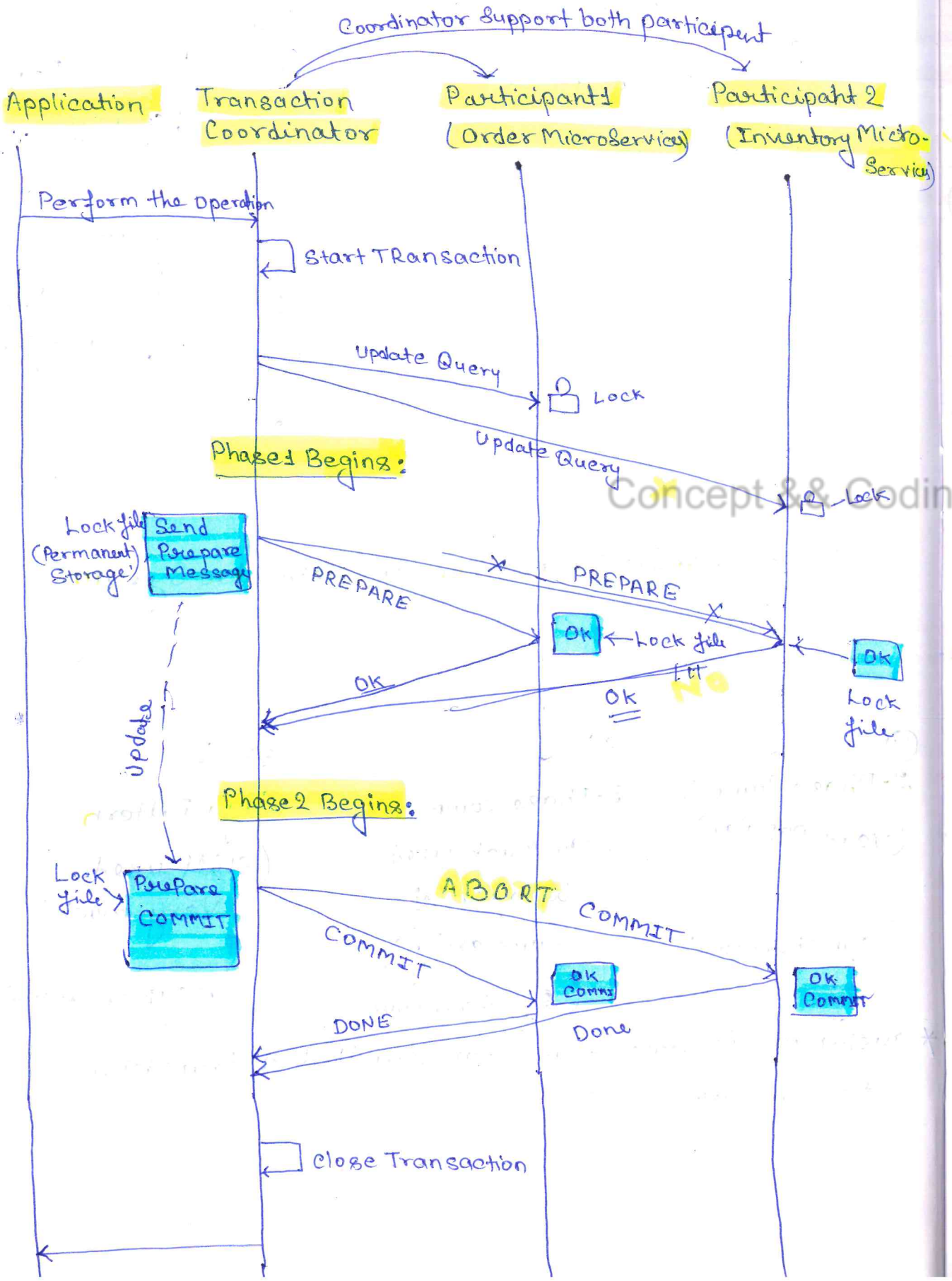
* customer is waiting you can say that a transaction to complete.

1: 2-Phase Commit :-

As name says, there are 2 phases in this protocal:

1: Voting or Prepare Phase

2 :- Decision or Commit Phase



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Interview Question

① What kind of issue can come?

1. Transation Coordinator can fail
2. Participant ~~can~~ get fail

1. Prepare MSG lost X
2. OK Message Lost X ~~Prepare message~~
3. COMMIT / ABORT Message is Lost.

Abort the Tx^n : Participant

② OK Message is Lost :- ~~Abort~~ Transaction Coordinator has wait some ~~of~~ timeout ~~can~~ is over. Abort the Tx^n. :TC.

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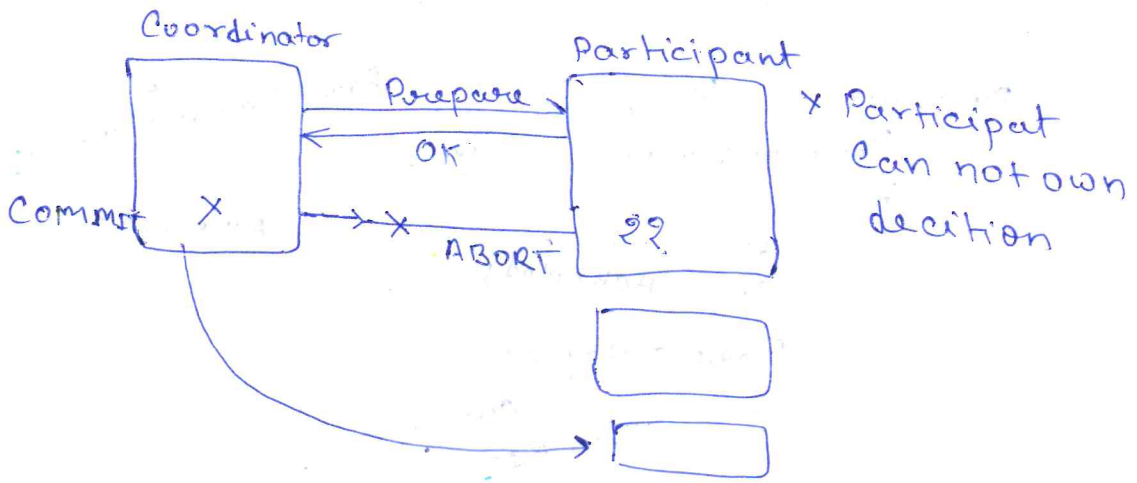
③ COMMIT/ABORT Message is Lost :-

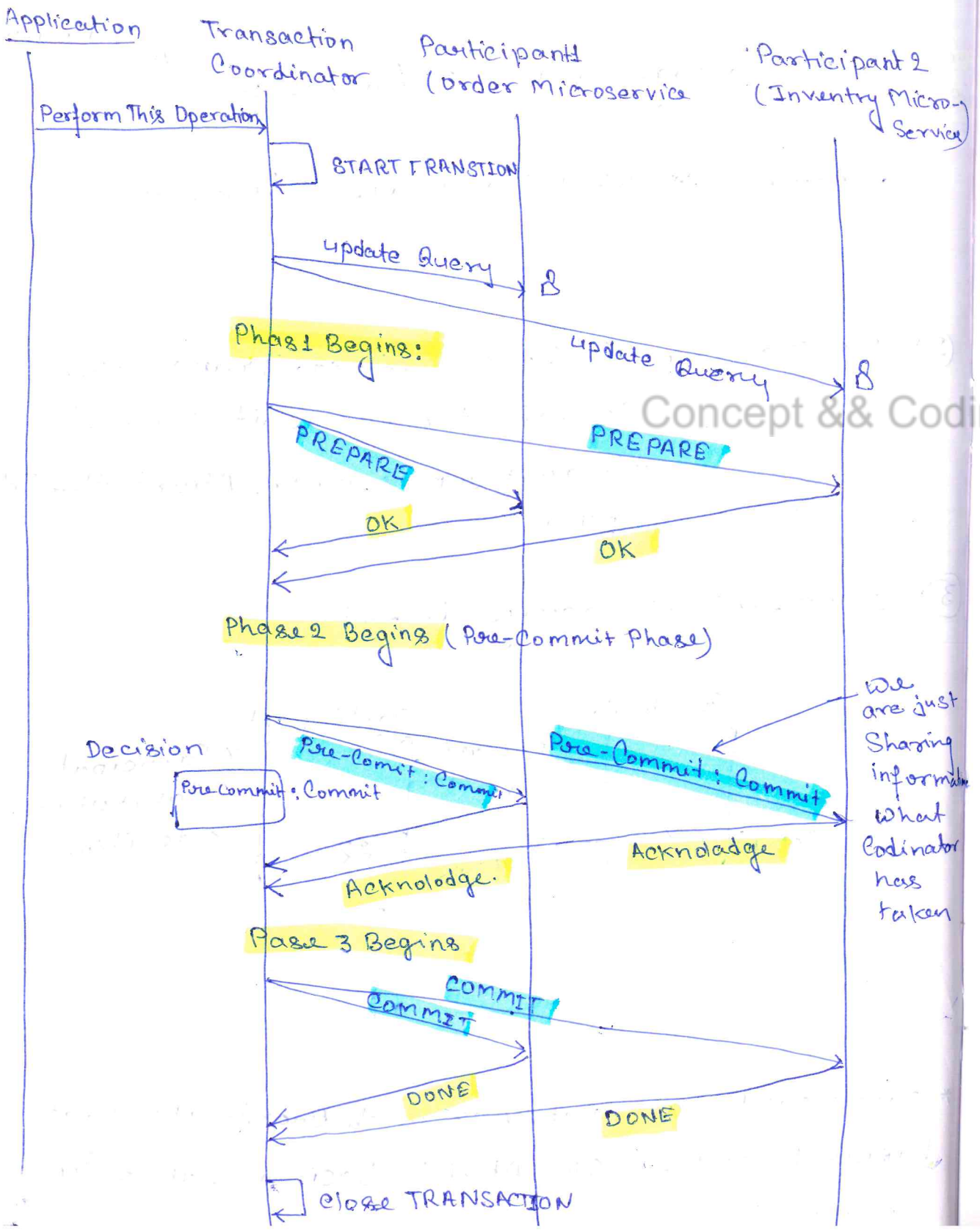


* participant has wait. Ask with Coordinator what happen? COMMIT OR ABORT. Desition has only Coordinator.

3-Phase Commit: (Non-Blocking Protocal) ⑥

As name says, there are 3 phases in this protocal

1. Prepare Phase [Same as 2PC]
2. Pre Commit Phase
3. Commit Phase



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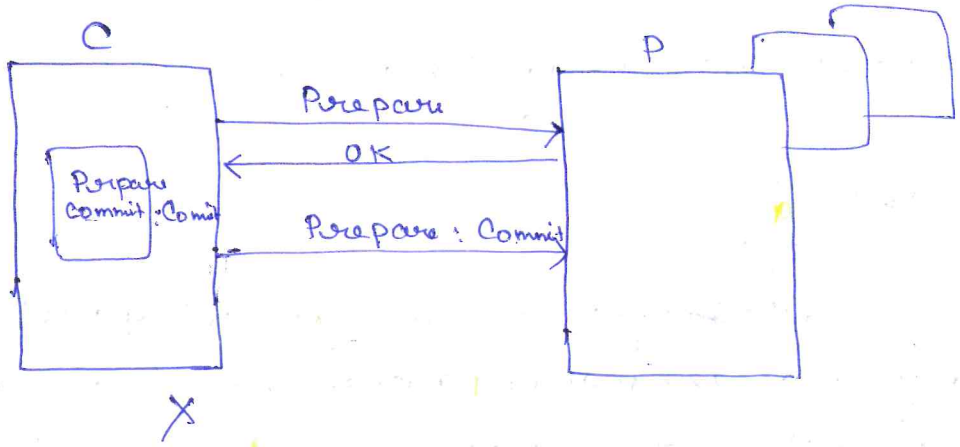
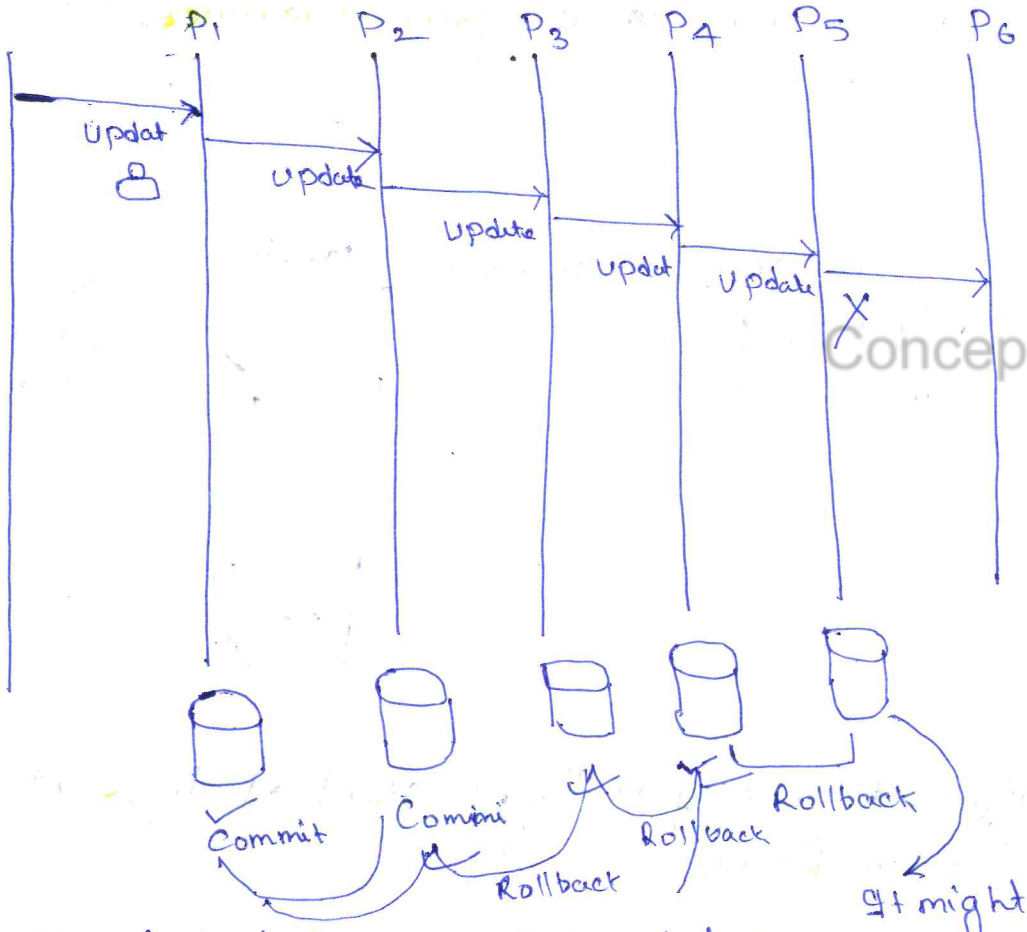
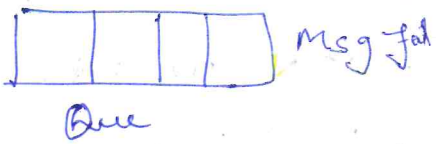


SAGA PATTERN (Asyncroumus)



* It kind of Sequential in Nature.

It might Pubblisien event



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